

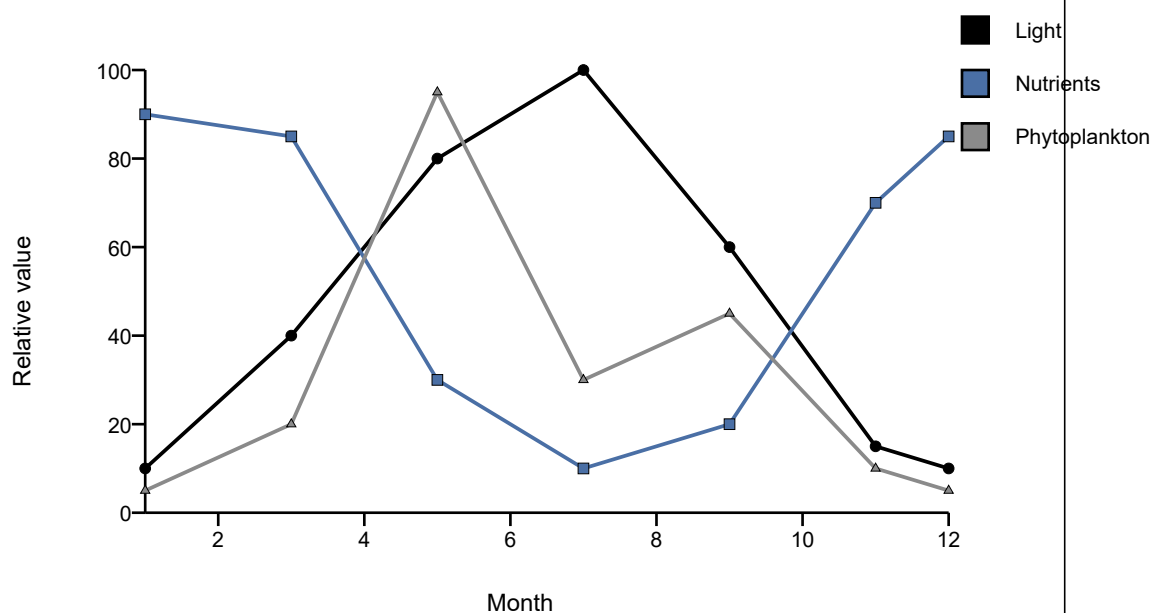
Questions by topic: Marine productivity & fishing - Set 1 (75 marks)

Answer all questions in the spaces provided.

0 1

Figure 1 shows the seasonal changes in light intensity, nutrient concentration and phytoplankton biomass in the surface water of a temperate ocean in the northern hemisphere.

Figure 1



0 1

1 Use Figure 1 to explain the spring peak and summer decline in phytoplankton biomass.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

0 1

2 Explain why tropical open oceans have low productivity all year despite high light and temperature.

[2 marks]

0 1 . 3

Explain why areas of upwelling and coastal waters are the most productive parts of the sea.

[3 marks]

0 1 . 4

Explain why phosphate is often the limiting nutrient in the sea.

[1 mark]

10

0 2

Table 1 shows features of some fishing methods.

0 2 . 1

Complete Table 1.

[6 marks]

Table 1

Method	Target species	One environmental disadvantage
Demersal trawling		
Purse seining		
Pelagic long-lining		

0	2	.	2
---	---	---	---

Explain why by-catch is a particular problem with shrimp trawling and drift netting.

[2 marks]

.....

.....

.....

0	2	.	3
---	---	---	---

Explain what is meant by **ghost fishing** and describe **two** ways it can be reduced.

[2 marks]

10

0 3

Table 1 shows life-history features of three fish species and the age at which they reach a catchable size.

Table 1

Species	Age at catchable size / years	Age at first breeding / years	Eggs per female per year	Maximum lifespan / years
Atlantic cod	2	4	Millions	25
Orange roughy	20	30	Thousands	150
Herring	2	2	Tens of thousands	20

0 3**1**

Use Table 1 to explain why orange roughy populations collapsed within ten years of large-scale fishing beginning in the late 1970s.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

0 3**2**

Explain why herring populations recover from overfishing much faster than cod.

[2 marks]

.....

.....

.....

.....

0 3**3**

Explain why the total mass of fish landed can stay the same while a population is being overfished.

[2 marks]

.....

.....

.....

0 3 . 4

Describe how the Greenland shark and blue (common) skate illustrate the threat of by-catch to k-selected species.

[2 marks]

10

0 4

The Russell formula describes changes in a fish stock: $S_2 = S_1 + (A + G) - (C + M)$, where S_1 and S_2 are the stock biomass at the start and end of the year, A = recruitment of young fish, G = growth, C = catch and M = natural mortality. Table 1 gives estimates for a cod stock.

Table 1

Term	Biomass / thousand tonnes
S_1	600
A	90
G	150
C	220
M	80

0 4 . 1

Calculate S_2 and state whether the stock is being fished sustainably.

[2 marks]

Answer

thousand tonnes

0 4 . 2

Calculate the maximum catch that would have kept the stock constant.

[1 mark]

Answer thousand tonnes

0 4 . 3

Explain why the catch is the only term in the formula that is easy to measure, and why the others are difficult.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 4 . 4

Describe how each of the following measures obtained from commercial catches can indicate overfishing.

[4 marks]

Catch per unit fishing effort

.....

.....

.....

Mean fish size and mean age

.....

.....

.....

10

0 5 . 1 Explain how each of the following reduces the environmental impact of fishing. **[8 marks]**

Square-mesh escape panels

.....

.....

.....

Minimum and maximum catchable sizes

.....

.....

.....

No-take zones

.....

.....

.....

Closed seasons

.....

.....

.....

0 5 . 2 Explain why catch quotas work well for pelagic species such as mackerel but poorly in mixed demersal fisheries.

[2 marks]

.....

.....

.....

.....

10

06 . 1

Describe **three** methods of reducing seabird by-catch in long-line fisheries.

[3 marks]

1

.....

2

.....

3

.....

06 . 2

Explain how restricting fishing effort and restricting fishing methods reduce overfishing.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

06 . 3

Describe how captive rearing and release ('population seeding') can increase wild populations of lobsters or oysters.

[2 marks]

.....

.....

.....

.....

06 . 4

Explain why collecting data on planktonic fish eggs and larvae helps to prevent population collapse.

[2 marks]

.....

.....

.....

.....

10

07 Table 1 shows the catch made by a shrimp trawler during one fishing trip.

Table 1

Component of catch	Mass / tonnes
Shrimps (target species)	0.6
Crabs, sea urchins, molluscs and starfish	5.9
Immature fish of commercial species	1.9

07 . **1** Calculate the percentage of the total catch that was by-catch.

[2 marks]

Answer %

07 . **2** Explain why shrimp trawling produces such a high proportion of by-catch.

[1 mark]

07 . **3** Explain why k-selected species such as the common skate and the Greenland shark are especially vulnerable to overfishing.

[3 marks]

07 . **4** Evaluate the methods that can be used to reduce the environmental impacts of fishing.

[9 marks]

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

.....
Extra space
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....

15

END OF QUESTIONS